SWEEP rule on the KD-Toric code

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Move this definition to the beginning of the paper.

Definition 0.1. We model the qubits on the finite toric with a side length n by the projection:

$$(x_1, x_2, .., x_d) \mapsto (x_1 \bmod n, x_2 \bmod n, .., x_d \bmod n)$$

Rewrite when we talk about the first time we have deviation clusters of size $\in (\alpha d, d)$. The reason is that having those assumptions means all the droplets in the past shrank; thus, we can use the ratio of edges to time-leaves in the big subgraph.

Claim 0.2. Assume that until step τ, there was no deviation cluster of size greater than d. Let $\hat{K}$ be a spanned slice induced by a deviation of a cluster state at step τ, and let ω be an explanation for $\hat{K}$ such that the number of vertices in ω is greater than βn. Then there is a subgraph (subtree) ω' of ω such that ω' has fewer than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves.

Proof. Let ω be an explanation of $\hat{K}$ with more than βn vertices. Since there is no deviation cluster of size greater than d until step τ, we have that every vertex of ω is a spanned slice induced by a boundary of a deviation cluster of size at most d. Thus, ω satisfies the shrink property in **??**, and by **??**, the ratio of edges to leaves of ω is at most γ. Observe that a spanning tree of ω has a ratio of edges to time-leaves that is at most ω's ratio, and the number of edges equals ω's vertices minus 1. Therefore, by applying **??** on ω's spanning tree, ω has a subtree with less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ time-leaves.

□

Proof. Now, since any two connected vertices are at a space-distance of at most 1, it follows that all the leaves in the subtree are at a space-distance of at most βn from each other. Thus, if $\beta < 1$, no two of them have the same space-coordinate modulo n. And therefore the probability to have such subtree is at most $p^{\frac{\beta}{4\gamma}n}$.

Because any ω with more than βn vertices implies the existence of a subtree with size less than $\frac{\beta}{2}n$, it's enough to bound the probability of having such a subtree. Considering that the trees are invariant to shifting by $n \cdot \mathbf{1}_i$, we have that it's sufficient to show that there are no such trees which are rooted in the finite cube $\mathbb{Z}_n^d \times \mathbb{Z}_t$. Finally, we use **??** to bound the number of such trees given a fixed root. Overall, we get:

$$\mathbf{Pr}\left[\text{No } \omega\text{s with more than } \beta n \text{ vertices}\right] \leq n^d t \left(\xi^2 p^{\frac{\beta}{4\gamma}}\right)^n$$

$\square$